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 Assistant Teacher Code : LOPF PANK THOS AJIA VLAD FOXT **TRIA**

Mathematical Methods 3/4

APPLICATION TASK 3

Question and Answer Book

The Adventures of the SAC-MAN 3



The Chairs Club Dating Game

Reading time is **15 Minutes**

Writing time is **120 Minutes**

Approved Materials:

- Any aids for the sketching of graphs
- A physical **OR** digital Mathematica Bound Reference
- **OR** A CAS Calculator with **physical** bound reference
- Stationary including pencils and a scientific calculator (if desired)

Materials supplied:

- Question and Answer book of 19 pages



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“Welcome to chairs club” - THOS walked in the room , yelling no contemplation

William Patel was bewildered at the scene. Now with Schmidt behind bars, he was living the life of a usual student.

Until...

He was made to participate in a game , Organized by the “Chairs Club” where he had to guess four “Criteria”, the JKSS Chairs club posited this game to test the power in William Patels’ heart

This adventure is about William Patel figuring out the four Criteria that the Chairs Club has made him to guess, however it will need some Mathematical Acumen, Indeed

The Criteria will now relate to a matchmaking service by the Chairs Club , this will be a unique experiment initiative, it’s not nefarious, right?



SECTION A: Please Read

- Answer **all** questions in the booklet
 - The JKSS Chairs Club has started an interesting initiative , the Dating game, the characteristics of this game as you may expect.
 - Your task is to help William Patel snap off all four of the challenge **Criteria** , using the power in his heart to play their sick game!
 - You must also help him survive the *Speed Dating Section*
 - **All persons included are included for Educational purposes only**
 - The sections in italics are story only and not required for the assessment
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Question 1 (12 Marks)

Immediately, after thinking “What would Kermond do?” , William begins analyzing the antics of a Chairs Club Executive *LOPF*

At mentor every school day *LOPF* likes playing Chess or Poker. If he plays Chess one mentor he has a **67%** chance of playing Poker the next mentor , if he plays Poker one mentor, he has a **54%** chance of playing chess the next mentor .

Suppose on Monday at mentor he plays Chess

a. Find the probability that

i. He plays Chess on Wednesday

(2 Marks)

ii . He plays Chess on Wednesday and Poker on Thursday

(2 marks)

William is still confused about the Criteria and analyzes someone else , *Ayan “Triple T”*

Triple T sometimes comes to school late. Suppose that if he comes on time to school on a particular day , the probably he comes late on the next day is p . However if he is late on a day, then the probability he comes on time on the next day is $0.6 - p$



Suppose on Monday he comes late to school

b.

i. Draw a Tree- diagram of what could happen on tuesday and wednesday

ii. Show that , in terms of p , the probability *Triple T* comes on time to school on Wednesday is $0.84 - 1.4p$ (2 Marks)

iii. State all possible value(s) of p (1 Mark)

Once again this analysis is not useful to find Criteria I, so he decides to analyze *THOS* himself ,

THOS decides to play with dice

He decides to roll two fair 6 sided die

Let A be the sum of digits from the face up on each roll

"I wONEder what could happen"

c. If $Pr(A > k) = \frac{26}{36}$, the possible value of k (2 Marks)

d. Determine the Expected Sum (2 Marks)



If this equation is less than zero for a closed finite non-zero interval, then he will be happy

e.

Determine the probability that *SRIK* is happy (3 Marks)

SD-HBA-GD

William has now figured out Criteria I , it's **to expect the unexpected!**

He has passed the first test

Now he is at Methods again, the Chairs Club have invited him to figure out Criteria 2



However tragedy has struck , he doesn't have a CHAIR!!!

Question 2 (10 Marks)

The **Chairs** Club messes with William's mind , and he must consider all students in his Methods class to sit in a **Chair**

It is known that SAC-MANs Methods Class Contains:

20 Year 12 students

16 Year 11 students (including himself)

a. The students arrange themselves in a straight line

i. In how many ways can the students be arranged in a straight line such that the Year 12 students and Year 11 students are grouped together? (1 Mark)

ii. In how many ways can the students be arranged in a straight line such that only Year 12s occupy the first 6 spaces (1 Mark)

iii. In how many ways can the students be arranged in a straight line such that either *Stephanie* (an Year 12) or *WONE* (An Year 11) occupy the first OR last space, but if one occupies either space the other cannot occupy the first or last space (1 Mark)

William is looking for a place to sit, it is known that the class is divided into tables of 6 chairs, only one student can sit in one chair in the table

b. Students are randomly selected to sit on the chairs, Determine the probability that:

i. There is atleast one Year 12 student in William's Table (1 Mark)

ii. there are at least two Year 12 students , but *Stephanie* and *Manal* (two Year 12's) must be sitting on Will's table?

(1 Mark)

iii. Either *Stephanie* **XOR** *Manal* must sit **next** to Will , but the other person must be on a different table (2 Marks)



The tables have a circular shape and are arranged in a peculiar grid such that the center of each table is 6 meters away from each other, William is seated at the exact center table of the grid, and each chair is evenly spaced around the circle, it is known one chair subtends an angle of 0 radians with the center on each table.

- c. Determine the probability that a randomly assigned student is more than 12 meters away from William (3 Marks)

William determines *Pickle* does not possess Criteria 2

It is known that 6 Year 12's, and 3 Year 11's (Other than William) possess Criteria 2

Assume that William had an equal chance of being placed in any table

- d. What is the probability, that William would sit with exactly 3 people possessing Criteria 2 on his table, given that at least one Year 12 is present in a **Chair** on his table (3 Marks)

Understanding this took a lot of power, so he deemed Criteria 2 to be to **possess the power in your heart**

Now that William has taken a seat in a **Chair**, The **Chairs Club** meeting can begin.



Question 3 (15 Marks)

"There are four criteria you must guess , I'm surprised you've snapped some off so quickly!" - THOS Exclaimed

Ayan "Triple T" was floating in the air listening intently, wondering if he was 4/4

LOPF has gathered a **sample** of people to determine who is in what Criteria

LOPF declares out of 1000 people he has surveyed , everyone has atleast one criteria

" $1000n + 0.9$ contain exactly one criteria"

" $1000n^2 - 800$ contain two criteria "

" 99 contain three criteria"

" $3n$ people are 4/4 on the Criteria List

Initially , William assumes that someone's criteria is independent of any other

a. The following table is a probability distribution of people with x amount of Criteria satisfied and s , amount of successful speed dates in the game ,

Fill it in the estimated probabilities in terms of n (1 Mark)

x	0	1	2	3	4
$Pr(X = x)$					
s	0	3	6	9	10

b.

i. State all possible equations that can be used to find the value of n (1 Mark)

ii. State the value of n , correct to four decimal places

(1 Mark)

SAC-MAN initially mainly speed dates those who satisfy a $\frac{3}{4}$ on the Criteria list.

c. Find the probability that a randomly selected person is a $\frac{3}{4}$, given that they have atleast 2 of the Criteria (2 Marks)



d . What is the expected number of successful speed dates someone in the survey would have had?
(2 Marks)

William randomly selects a k amount of people to find how many Criteria they follow
Let S be the random variable that denotes the amount of people who satisfy exactly 3 of the Criteria

e.

i If $k = 6$ state $Pr(S = 3)$

(1 Mark)

ii. For what value of k is $E(S)$ closest to 9

(1 Mark)

Consider the case where $k > 6$

William believes THOS is *in* if more than 6 people in the sample are $\frac{3}{4}$'s (But NOT $\frac{4}{4}$'s) , otherwise he's *out*

William knows that if the first three people are $\frac{3}{4}$'s then the probability THOS is *in* is 0.67

f. State the value of k

(2 Marks)

It turns out the initial Probability distribution was biased , for a large sample, the variance of the amount of $\frac{3}{4}$'s is 2 people

g. If $Pr(S = 6) = p$, state the range of p

(3 Marks)



h. If $p = 0.3$, find the mode of S in terms of k , Explain your answer
(2 Marks)

Now that William has analyzed the Criteria enough, the **Chairs Club** has sent him across to an isolated island in the Indian Ocean, in this island he must continue the next part of the Game in order to determine Criteria 3

Question 4 (18 Marks)

At the Island, William Patel must engage in a *Speed-Dating* Session, as organized by the Chairs Club. The time in minutes of a speed date T is a Random Variable. THOS designs the pairings to be modelled by the Probability Density Function:

$$f(t) = \begin{cases} \frac{kt}{t^2 + 1}, & 0 < t < b \\ 0, & \text{otherwise} \end{cases}$$

a. If William has a 60% Chance of having a *Date* greater than 5 minutes, state the value(s) of k and b , correct to three decimal places (2 Marks)

If the conversation greater than 3 minutes and less than 8 minutes, the *Date* is called *insightful*

b. Given that William has a *Date* greater than 5 minutes, find the Probability, correct to 4 decimal places it is *insightful*
(2 Marks)



The Three Chairs Club Executives are suddenly jealous of his success, and illegally enter the island themselves to self-insert themselves into the lore .

TRIA is the first executive to enter , he brings some paneer from the path of Shawarma hoping it would help him secure longer dates, the paneer is laced with a strange form Chemical X , treated with Ethanoic acid, to form a relaxation agent, call this chemical *Kermondium*

C the concentration of *Kermondium* in mol/L in the milk used to produce a random packet of Paneer is Normally distributed with a certain mean and standard deviation

The probability that concentration of *Kermondium* in a randomly select packet of paneer is under 10 mol/L certain a threshold concentration is the same as the probability that William will have a date under 10 minutes , it is also known that the most frequent concentration on average has the same frequency as William's most frequent Speed date time

c. State the Mean and Standard Deviation of C (2 Marks)

Because of his Paneer, *TRIA* has a better distribution of speed dates (in his opinion) Initially , before the paneer was produced , the distribution of times had a mean of k minutes and a standard deviation of 2 minutes

d.

i. find the value(s) of k such that $(4.6 \times 10)\%$ of his dates take between 5 and 9 minutes (3 Marks)



After he consumes the Paneer , the timing of speed dates becomes normally distributed with a mean of h and Variance of h^2

- ii. For any sample of speed dates, $n\%$ are feasible to occur in real life, state the value of n , correct to two decimal places (2 Marks)

LOPF now enters the island, the Time for his speed dates is a random variable T_2 Where

$$T_2 = 3T + 5$$

- e. Determine the probability density function of T_2 (2 Marks)

Let $h(t)$ be the probability density function of LOPF's Speed Date Times, such that the maximum value of the function is 99

- f. State the transformations from $g(t)$ to $h(t)$ (1 Mark)



Ultimately , it is proven that the executives are not able to beat William Patel, he is more in than any of them all. He unlocks all criteria , and now he must use that insight to crack the final password

Question 5 (10 Marks)

The hill to the final underground lair is modelled by the family of functions, William must travel to it

$$f_k: [-3, \infty) \rightarrow \mathbb{R}, f_k(x) = x^3 - kx^2 + 9kx + 1$$

Where $k \sim N(0, 10^2)$

- a. Find the probability correct to 3 decimal places, such that $f_k(x)$ has two stationary points
(2 Marks)

- b. Find the probability correct to 3 decimal places, that a non zero area bound by $f_k(x)$ and the tangent to $f_k(x)$ at $x = k$ exists
(2 Marks)

Consider the case where $k = 2$

Two x co-ordinates a and b are randomly selected , each of these points will lie on the graph of $f_k(x)$

It is known that

$$a \sim N(0, 10^2)$$

$$b \sim N(0, 20^2)$$



c. Find the probability, correct to three decimal places that:

i. $a = b$

(1 Mark)

ii. $f(a) = f(b)$

(1 Mark)

iii. $a < b$

(1 Mark)

iv. $f(a) < f(b)$

(1 Mark)

v. $|a - b| < 5$

(2 Marks)



Question 6 (3 Marks)

William has to answer one final question to leave the sham of a show

From the first 16 natural numbers two numbers a and b are selected without replacement

Determine the probability that $a - b \geq 10$

END OF QUESTION AND ANSWER BOOK

